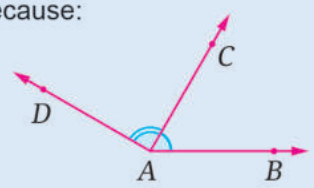
Relation between Angles

1 Adjacent Angles: They are two angles that lie in the same plane, share the same vertex, have a common side, and their outer sides lie on different sides of the common side.

Example:

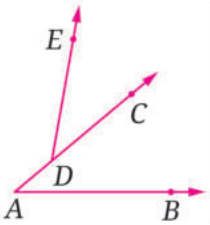
$\angle BAC$ and $\angle CAD$ are adjacent angles, because:

- They share the same vertex A, and have a common side $\overrightarrow{AC}$
- The outer sides $\overrightarrow{AD}$ and $\overrightarrow{AB}$ lie on different sides of the common side $\overrightarrow{AC}$.

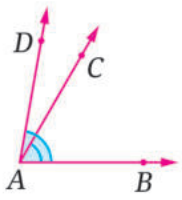


Think with your classmates

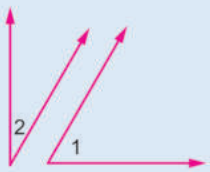
- Are $\angle BAC$ and $\angle CDE$ adjacent angles?

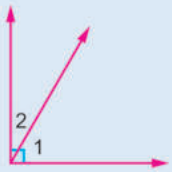


- Are $\angle BAD$ and $\angle BAC$ adjacent angles?



2 Complementary Angles: They are two angles whose measures have a sum of 90°.





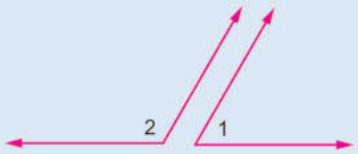
Example:

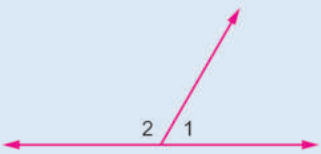
If $m(\angle 1) = 60°$, and $m(\angle 2) = 30°$

Then $\angle 1$ and $\angle 2$ are complementary angles, because:

$m(\angle 1) + m(\angle 2) = 60° + 30° = 90°$

3 Supplementary Angles: They are two angles whose measures have a sum of 180°.





Example:

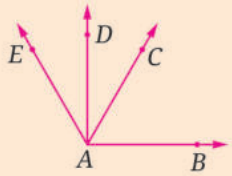
If $m(\angle 1) = 60°$, and $m(\angle 2) = 120°$

Then $\angle 1$ and $\angle 2$ are supplementary angles, because:

$m(\angle 1) + m(\angle 2) = 60° + 120° = 180°$

■ Critical Thinking

Write all the pairs of adjacent angles in the figure below.



■ Notation in Mathematics

The measure of an angle is denoted by the letter (m). Thus, the measure of angle (ABC) is written as follows: m ($\angle ABC$).

■ Think

Is it necessary for two angles to share the same vertex to be identified as complementary or supplementary angles?